

Layer 6: hierarchical priors, speciation, and memorization

A session report on belief propagation for diffusion scores

Session record — branch `claude/em-bp-denoiser-learning-e07ike`

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Abstract

This report documents one working session on the `GioviManto/diffusion` project. The session was directed by a single instruction: study two papers and take inspiration from them. Both turned out to be by the project’s advisors. The work that followed built a hierarchical (tree) prior with exact belief propagation and exact EM on it, derived the two diffusion time scales in closed form for that prior, and ran five experiments against them. It also produced two corrections to my own earlier claims, one of them substantive, and one negative result. Everything is on the feature branch; `main` is untouched. The test suite went from 50 to 101 tests, all passing.

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1 Scope, and how to read this

1.1 What the session was asked to do

The instruction was to study

- **P1** — Garnier-Brun, Mézard, Moscato, Saglietti, *How transformers learn structured data: insights from hierarchical filtering*, arXiv:2408.15138; ICML 2025 (PMLR 267:18831–18847);
- **P2** — Biroli, Bonnaire, de Bortoli, Mézard, *Dynamical regimes of diffusion models*, Nature Communications **15**, 9957 (2024); arXiv:2402.18491,

and continue the research. Marc Mézard is an author of both; Jérôme Garnier-Brun of the first.

1.2 Provenance of the source material — read this before citing

This matters enough to state before any result.

P1 has been read in full. Direct HTTP is blocked at this environment’s gateway: arxiv.org, www.nature.com, PMC, HAL and PMLR all return 403 on CONNECT, from both the fetch tool and curl. However the Hugging Face Hub exposes [hf://papers/2408.15138/paper.md](https://huggingface.co/papers/2408.15138/paper.md) through a server-side route, which returns the complete L^AT_EX rendering of the paper including all appendices. Every statement about P1 in this report is from that source.

P2 has *not* been read. It is not available on that route and every direct route is blocked. What is said about P2 here comes from web search — abstract-level statements about the two cross-overs and the entropic criterion — and **nothing mathematical is quoted from it**. The speciation criterion used throughout is derived from scratch in Section 4.1 in this package’s own conventions and checked against sampled trajectories. One search summary rendered the condition with a reciprocal, $t_S = \frac{1}{2} \log(1/\Lambda)$, contradicting its own statement that t_S diverges with dimension; deriving it independently avoided importing that slip. **P2 should be read before it is cited.**

A third paper matters and was not in the original list. Sclocchi, Favero and Wyart, *A phase transition in diffusion models reveals the hierarchical nature of data*, PNAS **122**(1) e2408799121 (2025), arXiv:2402.16991 — *cited by P1* — studies diffusion on hierarchically generated data directly. It is the closest prior work to Section 5.1 and is discussed in Section 6.2. It has not been read in full either; the account here is from search.

1.3 Structure

Section 2 summarises P1 as read. Section 3 describes the new code. Section 4 gives the derivations, all of which are verified numerically. Section 5 reports the experiments. Section 6 records what I got wrong and what came out negative. Section 7 states the limitations and what should happen next.

2 P1, as read

2.1 The data model

A root symbol x_0 is drawn from a vocabulary of size q . A transition *tensor* $\mathbf{M} \in \mathbb{R}_+^{q \times q \times q}$, with $M_{abc} = P(\text{children } (b, c) \mid \text{parent } a)$ and $\sum_{bc} M_{abc} = 1$, is applied for ℓ generations down a binary tree, producing 2^ℓ leaves. Entries are log-normally sampled and *non-overlapping*: if $M_{abc} > 0$ then $M_{a'bc} = 0$ for all $a' \neq a$. The production rules are therefore unambiguous, and at full hierarchy the whole tree including the root is deterministically recoverable from the leaves. The authors describe it as a simplified probabilistic context-free grammar with a fixed parse tree, and relate it to the Random Hierarchy Model of Cagnetta et al. (uniform rates, layer-dependent rules); the differences are log-normal versus uniform rates and a single tensor reused at every level.

2.2 Hierarchical filtering

This is P1’s central device and the part I originally implemented incorrectly (Section 6.1). At filter level k , the 2^k nodes at depth k are drawn **conditionally independently given the root**, each with the marginal the unfiltered model assigns it:

$$P(x_j = b \mid x_0 = a) = (\mathbf{p}_0 \mathbf{M}^{\sigma_0(j)} \mathbf{M}^{\sigma_1(j)} \dots \mathbf{M}^{\sigma_{k-1}(j)})_{ab}, \quad (1)$$

tracing the tensor along the unique root-to- j path, with $\sigma_m(j) \in \{L, R\}$ recording the branchings and $\mathbf{M}^L, \mathbf{M}^R$ the two marginalisations of $\mathbf{M}$. Levels $1, \dots, k-1$ are skipped. Below level k the ordinary process resumes.

The consequences are worth spelling out because they are exactly what makes filtering a clean experimental knob:

- correlations survive inside blocks of $2^{\ell-k}$ leaves;
- blocks remain correlated **through the root** — they are not independent;

- all marginals are preserved at every k ;
- at $k = \ell$ the leaves are conditionally i.i.d. given the root, and P1 notes that a Naive Bayes classifier is then optimal and every attention layer superfluous.

2.3 Inference and what it is used for

The graph is a tree at every k , so BP is exact and linear-time, converging in $2(\ell - k + 1)$ steps. This gives a *family* of exact algorithms BP_k , and running BP_k on data generated at a different filtering level is a well-defined *mismatched oracle*. That family is P1’s measuring instrument, and adopting it is the main methodological import of this session (Section 5.6).

2.4 Findings

1. Transformers match $\text{BP}_{k_{\text{train}}}$ ’s accuracy *and its calibrated probabilities*, in-sample and out-of-sample, including where that oracle is not optimal — evidence they implement the algorithm rather than merely scoring well.
2. **Sequential acquisition.** During training the predictions align first with BP_ℓ (leaf-to-root correlations only) and then with *decreasing* k : a clean staircase, shortest-range correlations first.
3. Attention maps develop blocks of size $\sim 2^{\ell-k}$, organised hierarchically across layers; layer-wise probes recover ancestors up to the depth of the probed layer.
4. Appendix F gives an explicit construction embedding BP in ℓ transformer layers, using $\mathcal{O}(q^2)$ memory slots per token to fold the downward pass into the upward one.
5. MLM pre-training sharply reduces the labelled data needed for root classification.

2.5 P2, as far as search reports it

The backward dynamics passes through two cross-overs. *Speciation*, where the coarse structure of the sample is decided by a symmetry-breaking mechanism, with a time scale set by the top eigenvalue of the data covariance — diverging logarithmically when that eigenvalue grows with dimension. *Collapse*, where trajectories are captured by a single training point through a mechanism analogous to glassy condensation, with a time scale set by an excess entropy, and the consequence that memorization is avoidable at finite times only if the dataset is exponentially large in the dimension. The collapse statement is about the *empirical score*: the exact score of the measure placing mass $1/N$ on each training point.

3 What was built

Two new modules, three new experiments, 51 new tests. Everything is pure `numpy`; no autodiff anywhere, consistent with the rest of the package.

3.1 `src/hierarchy.py`

The prior. `GaussianTree(depth, branching, rho, root_separation, filter_level)` is the continuous analogue of P1’s model:

$$z_{\text{root}} \sim \mathcal{N}(0, 1), \quad z_{\text{child}} = \rho z_{\text{parent}} + \sqrt{1 - \rho^2} \varepsilon, \quad (2)$$

with the leaves observed and internal nodes latent. Every node has unit marginal variance, matching the normalisation the chain priors use, so tree and chain families are directly comparable at a fixed noise schedule. For Gaussian innovations the leaf covariance is ultrametric,

$$\text{Cov}(\text{leaf}_i, \text{leaf}_j) = \rho^{2(L-d_{ij})}, \quad (3)$$

where d_{ij} is the depth of the lowest common ancestor.

Exact BP, twice.

- `tree_bp_gaussian` — information form (h, λ) , exact, $\mathcal{O}(n)$. Upward and downward updates touch only those two numbers per message, with nowhere for a projection step to hide. This is deliberate: the audit that preceded this work found the previous Gaussian BP was doing “something strange” instead of updating mean and variance.
- `tree_bp_grid` — grid messages, valid for *any* innovation law, using the same two primitives as the chain ($\mathbf{K}^\top f$ upward, $\mathbf{K}g$ downward) and the same `K[out, in]` convention. The leave-one-out sibling product is computed by prefix/suffix scans rather than by dividing out a message, since message entries are legitimately $\sim 10^{-16}$ in the tails.

EM on a tree. `tree_e_step` produces the **same** Ξ **matrix** as the chain E-step: $\Xi[k, j]$ is the posterior mass on the transition $u_j \rightarrow u_k$ summed over all tree edges. That is where the topology stops being visible, so *every* kernel in `src/kernels.py` consumes it unmodified — a chain-trained M-step and a tree-trained M-step are literally the same code. The one genuinely new piece of bookkeeping is the evidence: messages are renormalised at every node for numerical range, so $\log p(x)$ is recovered by accumulating the discarded log-scales. That is what keeps the monotone-ascent check available on trees.

Filtering. `filter_level=k` implements Section 2.2 faithfully; `filtered_tree_bp_gaussian` is the corresponding BP_k.

A bimodal root. `root_separation =  $\mu$` makes the root a symmetric two-component mixture $\frac{1}{2}\mathcal{N}(-\mu, 1 - \mu^2) + \frac{1}{2}\mathcal{N}(+\mu, 1 - \mu^2)$. Because this keeps unit variance, the leaf covariance — and therefore *every eigenvalue and every predicted time* — is bit-identical to the Gaussian-root tree. Only the modality changes. `tree_root_belief` returns the exact posterior of the root, which for a two-component prior *is* the class posterior.

3.2 `src/spectral.py`

The two time scales, in closed form and in this package’s conventions: the speciation crossover `speciation_time`, the measurable commitment, the AR(1) spectrum and its limit, the per-site excess entropy and the collapse dataset size.

3.3 Experiments

File	Parts	Subject
<code>exp_13_speciation_cascade.py</code>	<code>spectra, cascade, levels, ordering, block_independent, speciation</code>	the ladder, and what learns it
<code>exp_14_memorization_collapse.py</code>	<code>budget, collapse, time</code>	memorization and its time scale
<code>exp_15_bp_oracles.py</code>	<code>ladder, alignment, mismatch</code>	the mismatched-oracle probe

3.4 Validation

Every load-bearing object is computed twice, by independent routes:

Quantity	Checked against	Agreement
Analytic ultrametric spectrum	eigh of the dense covariance	10^{-15}
Information-form tree BP	dense linear solve	10^{-10}
Grid tree BP	information form	2×10^{-6}
Filtered BP (BP_k)	dense linear solve	10^{-15}
Tree E-step evidence	closed-form Gaussian likelihood	rel. 10^{-6}
Root belief	dense linear solve	10^{-5}
<code>commitment</code> (t, Λ)	40 000 sampled trajectories	< 0.02
Ξ mass	edge count	exact
EM monotonicity	—	violation = 0

4 Derivations

4.1 The speciation crossover

Along an eigendirection of the clean covariance with eigenvalue Λ , the variance-preserving forward process splits into signal and noise:

$$\text{Var}(\langle v, x_t \rangle) = \alpha_t^2 \Lambda + \Delta_t, \quad \alpha_t^2 = e^{-2t}, \quad \Delta_t = 1 - e^{-2t}. \quad (4)$$

The mode is decided while signal dominates and undecided once noise does, so the crossover sits at $\alpha_t^2 \Lambda = \Delta_t$:

$$\boxed{t_S(\Lambda) = \frac{1}{2} \log(1 + \Lambda)} \quad (5)$$

which reduces to the familiar $\frac{1}{2} \log \Lambda$ for large Λ . The measurable form is the correlation between the projection at time t and at time 0,

$$\text{commitment}(t, \Lambda) = \frac{\alpha_t \sqrt{\Lambda}}{\sqrt{\alpha_t^2 \Lambda + \Delta_t}}, \quad (6)$$

which passes through $1/\sqrt{2}$ exactly at t_S . Verified analytically and against 40 000 sampled forward trajectories (agreement < 0.02).

Remark 1 (A distinction not to blur). In P2, speciation is a *symmetry breaking* and needs a multi-modal distribution to have something to choose between. A Gaussian prior has no class to speciate into, so what `commitment` locates is an *information crossover* sharing the same criterion. Section 5.2 closes that gap by making only the root bimodal, with the covariance held bit-identical.

4.2 The chain has a bounded spectrum; the tree does not

For a stationary AR(1) chain with $\text{Cov} = \rho^{|i-j|}$, the spectrum is bounded at every length by the spectral density at zero frequency:

$$\Lambda_{\max} \longrightarrow \frac{1 + \rho}{1 - \rho}. \quad (7)$$

So t_S saturates. This is not a technicality: it says the chain family studied in Layers 1–5 sits outside the regime P2 describes for image data, where Λ grows with dimension and t_S diverges logarithmically.

For the balanced tree, writing S_d for the sum of covariances over a depth- d subtree,

$$S_d = 1 + (b-1) \sum_{m=1}^{L-d} b^{m-1} \rho^{2m}, \quad \Lambda_d = S_{d+1} - b^{L-d-1} \rho^{2(L-d)}, \quad (8)$$

the latter with multiplicity $(b-1)b^d$, plus the uniform mode at S_0 . Exactly $L+1$ distinct eigenvalues, hence $L+1$ speciation times. When $b\rho^2 > 1$ the top eigenvalue grows geometrically in depth, i.e. logarithmically in dimension.

4.3 Filtering removes rungs from the ladder

This is where P1’s knob and P2’s time scale meet in one formula. Under the correct filtering the leaf covariance is block-diagonal-plus-constant: the depth- $(L-k)$ subtree covariance inside each of the b^k blocks, and ρ^{2L} between them. Its eigenvectors are

- (i) within-block contrasts — the subtree’s own non-uniform modes, each multiplicity multiplied by b^k ;
- (ii) between-block contrasts, constant on each block and summing to zero across blocks, with the single eigenvalue $\Lambda_{\text{between}} = S_0^{(L-k)} - b^{L-k} \rho^{2L}$ and multiplicity $b^k - 1$;
- (iii) the uniform mode, $S_0^{(L-k)} + (b^k - 1)b^{L-k} \rho^{2L}$.

So the top k rungs **merge into one**, and the number of distinct speciation times falls from $L+1$ to $L-k+2$, reaching 2 at $k=L$ where the covariance is equicorrelated. Measured rung counts confirm this exactly (Section 5.6).

4.4 A closed-form collapse criterion

For a Gaussian AR(1) chain the differential entropy is exactly extensive, so the per-site excess entropy relative to the terminal measure $\mathcal{N}(0, 1)$ is

$$s = -\frac{1}{2} \log(1 - \rho^2), \quad (9)$$

exactly, with no fitting, and the dataset size below which memorization is forced is e^{ns} . At $\rho = 0.85$ that is 0.641 nats per site. At finite t the noised law is $\mathcal{N}(0, \alpha_t^2 C + \Delta_t I)$, so

$$s(t) = -\frac{1}{2n} \log \det(\alpha_t^2 C + \Delta_t I), \quad (10)$$

and $n s(t_C) = \log N$ gives a **predicted collapse time with no fitted constants**.

5 Results

5.1 The speciation ladder, measured

Depth-5 binary tree, $\rho = 0.9$, 32 leaves. Predicted t_S against the measured crossing of the commitment curve through $1/\sqrt{2}$, along the forward process and along the reverse SDE driven by the exact tree-BP score:

Level	Λ	t_S predicted	forward	reverse SDE
−1 (whole tree)	14.271	1.363	1.368	1.336
0	3.113	0.707	0.732	0.725
1	1.804	0.516	0.521	0.520
2	0.996	0.346	0.340	0.358
3	0.498	0.202	0.208	0.222
4 (sibling leaves)	0.190	0.087	0.087	0.125

Six distinct transitions spanning $15.7\times$ in time *within one dataset*. Pearson $r = 0.9998$ for both columns; forward crossings agree to $\leq 3.5\%$ at every level. The reverse column’s deviations are $-2, +3, +1, +4, +10, +43$ percent from coarsest to finest — the monotone growth toward the fine end identifies this as integrator error rather than a failure of the prediction, since the finest level speciates at $t = 0.087$, within a factor of four of the sampler’s $t_{\min} = 0.02$. Four of six levels agree to $\leq 4\%$.

The chain has no ladder. Same analysis on the AR(1) chain:

ρ	top eigenvalue, $n = 8 \rightarrow 512$	limit	t_S saturates at
0.50	$2.57 \rightarrow 3.00$	3.00	0.693
0.85	$5.49 \rightarrow 12.32$	12.33	1.295
0.95	$7.03 \rightarrow 38.52$	39.00	1.844

Sixty-four times the length moves the top mode by under $2\times$. A tree at $\rho = 0.9$ instead runs $3.12 \rightarrow 23.31$ over depths 2–6, its levels spanning $t_S \in [0.087, 1.595]$. **This bounds how far Layers 1–5 generalise toward image-like data**, and is the least comfortable finding of the session.

5.2 Genuine symmetry breaking

Depth-4 tree, $\rho = 0.9$, $\mu = 0.9$, 256 paths, t_S predicted 1.136. The order parameter is the exact class posterior $P(\text{root} > 0 \mid x_t)$, which BP returns from the upward pass alone; using the sign of the tree-mean instead would conflate the class with the within-class fluctuation.

Root	correlation crossing	class-choice time	final magnetization
two-component	1.155 (+1.7%)	1.265	0.828
Gaussian	1.031 (−9.2%)	0.940	0.647

Two readings. First, the covariance-set crossing is where it should be for both roots — 1.155 and 1.031 bracket the predicted 1.136, within the $\pm 10\%$ the integrator costs at this resolution. **Adding modality does not move it**, which is the control working. Second, the class choice for the two-component root happens at $t = 1.265$, *earlier in the generation* than the information crossover at 1.136, and 35% earlier than the corresponding time for the Gaussian root: two well-separated classes can be told apart while noise still swamps the within-class detail.

Remark 2 (Precision, stated against my own result). With 256 paths the binomial standard error on the agreement curve is 0.031, and the integrator contributes the $\pm 10\%$ above. The 1.265 versus 0.940 gap is far outside both; the +1.7% agreement of the bimodal correlation crossing is *inside* them and must not be read as better than the −9.2%.

5.3 Where each method’s error lives in the hierarchy

Depth-4 tree, $\rho = 0.9$, $N = 256$ trees, both network parameterizations trained. EM recovered $\hat{\rho} = 0.9161$ (true 0.9) and $\hat{q} = 0.1773$ (true 0.19) in 150 iterations with monotonicity violation exactly 0.

t	EM-BP	best network	ratio
0.1	0.0098	0.1852 (ε)	18.9×
0.2	0.0125	0.2245 (ε)	18.0×
0.4	0.0167	0.2489 (x_0)	14.9×
0.8	0.0334	0.2587 (x_0)	7.8×
1.6	0.0815	0.3332 (x_0)	4.1×

Grid tree BP given the true kernel scores 0.0000 at every t , which is the validation the rest of the part rests on.

More interesting is *where* the error sits. The right null is a method whose per-mode error is uniform, which would put a share of the total squared error on each level proportional to its multiplicity (1, 1, 2, 4, 8 out of 16 here). Dividing the measured share by that null gives a concentration ratio, whose spread across levels says how much a method’s error knows about the hierarchy:

Method	spread of concentration ratio (min $\rightarrow$ max over t)
network, ε -prediction	1.6 × – 2.2 ×
network, x_0 -prediction	2.8 × – 12.6 ×
EM-BP	224 × – 6957 ×

The ε -network distributes its error uniformly over the hierarchy at every noise level, as if the levels did not exist. EM-BP’s error is concentrated by two to four orders of magnitude, and it *migrates*: at $t = 0.1$ it sits on the finest levels, and by $t = 1.6$ all of it is on the single coarsest mode — at large t only the top mode survives the noise, so the kernel misestimate has nowhere else to appear. The x_0 -network is genuinely intermediate, and is reported as such because it is the better parameterization at large t .

5.4 Memorization: the axis a BP score does not have

The collapse mechanism is a statement about the *empirical* score, whose sufficient statistic is the training set itself: $N \times n$ numbers, growing in both dataset size and dimension. A BP score’s sufficient statistic is the fitted kernel — two numbers for a Gaussian AR(1) — reached only through Ξ , which is an average, and *independent of n* . So the prediction is not that BP memorizes less but that **the axis does not exist for it**.

Measured as nearest-training-chain distance of generated samples divided by the same quantity for a fresh draw from the true prior (1.0 = no memorization, 0 = full collapse), $\rho = 0.85$:

n	N	wall e^{ns}	empirical	DSM (ε)	EM-BP	true-prior BP
8	64	88.8	0.457	0.800	1.079	1.076
8	256	88.8	0.540	0.929	1.058	1.044
8	1024	88.8	0.660	0.980	1.015	1.012
16	64	1.5×10^4	0.332	0.751	0.972	0.992
16	256	1.5×10^4	0.387	0.906	1.040	1.026
16	1024	1.5×10^4	0.456	1.066	1.066	1.072
32	64	4.3×10^8	0.268	0.790	1.021	1.024
32	256	4.3×10^8	0.308	1.053	1.039	1.024
32	1024	4.3×10^8	0.335	1.115	1.023	1.025

Both dependences come out in the predicted direction. At fixed N the empirical score memorizes *more* as the chain gets longer ($0.457 \rightarrow 0.332 \rightarrow 0.268$ at $N = 64$) — the curse of dimensionality; at fixed n it recovers as N grows. **EM-BP stays in $[0.97, 1.08]$ in every one of the nine cells**, agreeing with the true-prior reference to ≤ 0.03 : its behaviour does not depend on n or N at all.

Remark 3 (The DSM column must be read with its standard deviation). At $n = 32$ the network’s sample standard deviation is 1.19–1.27, i.e. over-dispersed by 19–27%, so its ratios above 1 there are inflated by generating too broadly rather than by generalizing better. At $n = 8$ and 16, where its dispersion is closer to correct, its ratios sit below BP’s. This is why `sample_std` and `lag1_corr` are logged in every row.

This is the mechanism behind Layer 5’s headline. The $\geq 64\times$ data advantage and the $N^{-1/2}$ rate had been reported as empirical facts. This is why they hold.

5.5 The collapse time, predicted

The criterion $ns(t_C) = \log N$, with no fitted constants, was tested over

$$n \in \{8, 16, 32\} \quad \times \quad N \in \{32, 128, 512, 2048\}.$$

- **Pearson** $r = +0.990$ over the 9 settings where a finite-time collapse is predicted ($+0.995$ against the same measurement made on fresh rather than training chains).
- The relation is **affine, not proportional**: $\text{measured} = 1.716 \times \text{predicted} + 0.276$, residual s.d. 0.029. Forcing it through the origin gives residual s.d. 0.131, $4.5\times$ worse.
- Independently, $dt_C/d\log N$ is -0.039 predicted versus -0.061 measured at $n = 16$, and -0.051 versus -0.079 at $n = 32$ — *the same $1.55\times$ slope factor at both chain lengths*.

The criterion gets the ordering, both dependences, and the N -scaling to within a constant factor; it does not get the absolute time. That is what should be expected of a leading-order statement combined with an arbitrary “half of $\log N$ ” landmark, and the offset is reported rather than absorbed into a fitted constant.

Where it does not apply. At $n = 8$ with $N \geq 128$ the criterion returns *no finite-time collapse* (total excess entropy 4.487 nats against $\log 128 = 4.852$), yet collapse is measured at $t = 0.234, 0.170, 0.122$. There is no contradiction: the criterion is asymptotic in n , and a finite system always collapses eventually as $t \rightarrow 0$. Those three rows are excluded from the correlation and listed rather than dropped.

5.6 The mismatched-oracle probe

This is P1’s actual instrument, and adopting it is the main methodological result of the session.

The ladder collapses as derived. Rung counts at $\rho = 0.9$:

	$k = 0$	1	2	3	4	5	6
$L = 4$: distinct rungs	5	5	4	3	2	—	—
$L = 6$: distinct rungs	7	7	6	5	4	3	2
$L = 6$: ladder span	18.3	18.3	18.1	17.8	17.5	17.3	5.5

Exactly $L - k + 2$, as derived in Section 4.3. That $k = 0$ and $k = 1$ coincide independently reproduces P1’s remark that those two cases share a tree topology and differ only in the top transition probabilities.

The network implements the oracle matched to its training data. Train a denoiser on data filtered at k_{train} , test on *unfiltered* data, and ask which BP_k its posterior mean is closest to:

k_{train}	arg min $_k$ at $t = 0.1/0.2/0.4/0.8$	margin over runner-up
0	0 / 0 / 0 / 0	—
1	0 / 0 / 0 / 0	<i>exactly degenerate</i>
2	2 / 2 / 2 / 2	+1.8/ + 4.1/ + 4.3/ + 1.6 %
3	3 / 3 / 3 / 3	+17.8/ + 29.0/ + 24.3/ + 5.9 %
4	4 / 4 / 4 / 4	+63.3/ + 60.4/ + 35.3/ + 6.2 %

16 of 16 cells pick the oracle matched to the *training* distribution rather than the one matched to the test data, with the margin growing in k_{train} . This is P1’s Figure 2 finding, transplanted to diffusion. Two exclusions, both explained rather than hand-waved:

- $k_{\text{train}} = 1$ **is exactly degenerate with 0 in a Gaussian tree** — the distances agree to five decimals, gap 0.00%. This is a real limitation of the Gaussian analogue: P1’s transition *tensor* can correlate the two children given the parent, whereas a linear-Gaussian edge makes siblings conditionally independent by construction. The one case where their filtering alters probabilities without altering topology is *unrepresentable here*.
- **At $t = 1.6$ the arg min stops measuring alignment** — not because the oracles converge (their spread is still 0.22) but because the network’s distance to *every* oracle is ~ 1.50 , seven times that spread. Every row carries `oracle_spread` so this is checkable.

5.7 The replicated architecture comparison

A job pending from before the session completed during it, and it moves a headline. Four seeds, oracle over both receptive-field radius and parameterization for both networks, mean relative score error over five noise levels, $\pm$ standard error over seeds:

N	EM-BP (13 params)	global MLP (25,505)	local CNN, oracle r (6,081)
128	0.0368 $\pm$ 0.0067	0.3595 $\pm$ 0.0044	0.0774 $\pm$ 0.0024
512	0.0174 $\pm$ 0.0017	0.1794 $\pm$ 0.0035	0.0570 $\pm$ 0.0007
2048	0.0123 $\pm$ 0.0009	0.1246 $\pm$ 0.0027	0.0505 $\pm$ 0.0015
<i>Ratio to EM-BP</i>			
128	—	9.76 $\pm$ 1.79	2.10 $\pm$ 0.39
512	—	10.29 $\pm$ 1.02	3.27 $\pm$ 0.32
2048	—	10.15 $\pm$ 0.75	4.11 $\pm$ 0.32

Two things, one in each direction. The structured architecture really does close most of the gap: against the vanilla MLP the margin is $\sim 10\times$ and flat in N ; against the locality-respecting CNN it is $2\text{--}4\times$, so roughly 60% of the vanilla deficit was the architecture rather than the estimator. **But the remaining margin grows with data rather than closing** — $2.10 \rightarrow 3.27 \rightarrow 4.11$, well outside the standard errors — because EM-BP keeps paying down parametric error (a factor 3.0 over $16\times$ data) while the CNN flattens toward an approximation floor (a factor 1.5). “More data will close the gap” is the natural guess and it is wrong here.

This supersedes the single-seed figures previously recorded ($3.3\times$ at fixed $r = 6$, $2.75\times$ with an oracle over r , both at $N = 1024$).

6 Corrections, and what came out negative

A research record is worth less than nothing if it only contains what worked.

6.1 I had P1’s filtering construction wrong

Before reading the paper I implemented “filter at level k ” as b^k **independent** subtrees. P1 draws the depth- k nodes conditionally independently *given the root*, so blocks remain correlated through it at ρ^{2L} ; independent blocks put a zero there instead. The implementation now follows the paper, and a test pins the distinction rather than leaving it to a comment. The old sweep survives, renamed `block_independent` in code, docs and output filenames, because it measures a harsher truncation that is still well-defined — but calling it filtering was a misattribution, not a wording problem.

6.2 I claimed an intersection was empty when it is not

An earlier version of the study note said the hierarchy- $\times$ -diffusion intersection was “untouched by either paper”. Sclocchi, Favero and Wyart (PNAS 2025), *cited by P1*, study exactly that and find a phase transition at which reconstruction of high-level features drops sharply while low-level features evolve smoothly. The claim is withdrawn. What the Gaussian setting adds is more modest and more useful:

1. the whole ladder is analytic, so measurements are calibration against a known answer rather than discovery;
2. filtering acts on the ladder in a computable way (Section 4.3);
3. their sharp/smooth split is *reproduced* here — smooth crossovers at every Gaussian level, genuine symmetry breaking only once the root is made bimodal, with the covariance held bit-identical (Section 5.2).

6.3 The negative result: no sequential acquisition

P1’s headline dynamical finding does not reproduce here. Training on unfiltered data and tracking which BP_k the denoiser is closest to, the argmin does move in the predicted direction at two of five noise levels — $2 \rightarrow 2 \rightarrow 0$ at $t = 0.4$ and $2 \rightarrow 1 \rightarrow 1$ at $t = 0.8$ over the first ~ 1000 gradient steps — and then **saturates for the remaining 31,000**. At $t = 0.1, 0.2$ and 1.6 it never leaves $k = 0$. Nothing like their staircase.

Three plausible reasons, none verified: their oracles are far more distinguishable (a discrete non-overlapping tensor versus a Gaussian edge — see the $k=0/k=1$ degeneracy); a denoising regression loss converges in far fewer steps than MLM, which needs $\sim 10^3$ epochs; and the architecture and task differ. The honest summary is that **this setting is probably too easy to resolve the phenomenon**, which is a different statement from the phenomenon being absent.

The earlier probe (`exp_13 ordering`: resolve the error onto levels and watch it move) found nothing at all, and is retained only as a record of a measurement that could not have answered the question.

6.4 A non-monotonicity that is not a bug

EM’s *error against the truth* is non-monotone in iteration count even though its log-likelihood is monotone: total relative error $0.0729 \rightarrow 0.0083 \rightarrow 0.0162$ at 32/64/128 iterations, with $\hat{\rho} = 0.8708 \rightarrow 0.9068 \rightarrow 0.9158$ against a true 0.9. Nothing is wrong. EM converges to the *maximum likelihood estimate*, which at $N = 256$ is $\hat{\rho} = 0.9161$, not to the true parameter; the trajectory merely passes close to 0.9 on the way. Reading the minimum at 64 iterations as “the right stopping point” would be fitting the answer. The monotonicity violation is exactly 0 throughout, which is the quantity that would actually signal a broken M-step.

6.5 Tree EM converges, but slowly enough to look broken

$\hat{\rho} = 0.7360 \rightarrow 0.7483 \rightarrow 0.7487$ at 50/100/150 iterations (true 0.75), with zero monotonicity violation throughout, and $\hat{q} = 0.4418$ against 0.4375. Internal nodes are never observed, so the missing-information fraction is far larger than on a chain and EM’s linear rate is correspondingly slower. An estimate read at 40 iterations is off by 0.06 and reads exactly like a broken M-step. This cost real time to diagnose and is now asserted as a *rate* property, at two budgets, in the test suite.

6.6 A confounded sweep, reported as confounded

The `block_independent` sweep shows the EM-BP advantage running 29–99 $\times$ and not shrinking as correlations reach further, while the network’s error is flat in k ($0.414 \rightarrow 0.347$). But fixing the number of *sequences* does not fix the number of *edges* (16, 24, 28, 30 for $k = 1 \dots 4$), so larger k hands EM more data at the same nominal budget. The sweep moves two things at once, so the *rising* advantage is not attributable to correlation range. Only the negative is safe.

7 Limitations and what should happen next

7.1 Limitations

1. **P2 is unread.** Nothing here depends on its notation, but a write-up that cites it must use its symbols, and that correspondence has not been checked against the source.

2. **The Gaussian tree cannot represent sibling correlation given the parent**, so P1's $k=0$ versus $k=1$ distinction is unrepresentable. A discrete-tensor tree would be needed.
3. **The chain family is outside P2's regime**. Its top eigenvalue is bounded, so nothing about diverging speciation times transfers from Layers 1–5 to image-like data without crossing that gap explicitly.
4. **The collapse result uses a two-parameter kernel**. The obvious objection is that such a kernel *cannot* memorize, so the result is trivial. `exp_14 -set include_mdn=True` runs the same comparison with a mixture-density kernel of several hundred parameters and the same n -independent statistic; it has not been run.
5. **Inference cost remains the honest loss** of the whole programme: grid BP is $\mathcal{O}(nM^2)$ per evaluation against a few matmuls for a forward pass, and reverse diffusion calls the denoiser at every step.

7.2 Next, in priority order

1. Read P2, and read Sclocchi/Favero/Wyart — the closest prior work, whose territory an earlier version of my notes claimed was empty.
2. The nonparametric collapse test (limitation 4), which separates the mechanism from the parameter count.
3. A **discrete-alphabet tree**: closest to P1's actual data model, with exact vector messages and no closure. `src/discrete.py` does chains and `src/hierarchy.py` does continuous trees; the two have not been crossed. This would also make the $k=0/k=1$ case measurable.
4. The speciation cascade under a *learned* score rather than the exact one: which levels of the ladder each method gets right dynamically.
5. A non-Gaussian tree. `tree_bp_grid` and `fit_em_tree` already support it with no new machinery.

8 Reproducibility

8.1 Running it

```
cd research/nongaussian-bp
../../.venv/bin/python -m pytest tests/ -q      # 101 passed, ~3.3 min

python experiments/exp_13_speciation_cascade.py --list-parts
python experiments/exp_13_speciation_cascade.py --only cascade
python experiments/exp_14_memorization_collapse.py --only collapse
python experiments/exp_15_bp_oracles.py
```

Add `-quick` to any experiment for a minutes-scale smoke run. `-set KEY=VALUE` overrides any key printed in `params_*.json`; unknown keys are rejected, so a typo in a job script fails immediately rather than silently running defaults. `hpc/slurm_layer6.sbatch` is an 8-task SLURM array covering the parts worth a cluster.

8.2 Where things are

Path	Contents
docs/PAPER_CONNECTIONS.md	the study note; §0 records provenance
docs/EM_BP_LEARNING_COMPENDIUM.md	all results, §4.13 is Layer 6
docs/AGENT_HANDOFF_EM_BP.md	how to continue safely; 10 traps
questions/QUESTIONS_AND_ANSWERS.md	the question ledger, §[P] is Layer 6
research/nongaussian-bp/src/hierarchy.py	tree prior, tree BP, tree EM
research/nongaussian-bp/src/spectral.py	the two time scales
research/nongaussian-bp/experiments/exp_1{3,4,5}*.py	the experiments
research/nongaussian-bp/outputs/	committed CSV + JSON + PNG
research/nongaussian-bp/tests/test_hierarchy.py	51 tests

8.3 Conventions that must not be broken

Forward process $dX = -X dt + \sqrt{2}dW$, so $\alpha_t = e^{-t}$ and $\Delta_t = 1 - e^{-2t}$. Score identity $s(x, t) = -(x - \alpha_t \mathbb{E}[a \mid x])/\Delta_t$; every estimator goes through it, which is why `identity_residual` appears in every results CSV and must sit at machine precision. All randomness flows through `src.utils.rng_for`, which is reproducible *across processes* — it previously was not, and no experiment in the package was bit-reproducible as a result. Ξ uses `K[out, in]` and sums to the edge count.

Summary of the session

1. Identified both papers as the advisors' own; read P1 in full through a server-side route after direct access was blocked.
2. Built a hierarchical tree prior with a closed-form ultrametric spectrum, exact BP twice over, exact EM producing the same Ξ as the chain, and the two diffusion time scales in closed form.
3. Measured a ladder of six speciation transitions spanning $15.7\times$ in time, each within 3.5% of prediction ($r = 0.9998$).
4. Established that the AR(1) chain of Layers 1–5 has no such ladder, bounding how far the project generalises.
5. Showed the memorization axis does not exist for a BP score, supplying the *mechanism* behind Layer 5's headline sample-efficiency result.
6. Verified a closed-form collapse time at $r = 0.990$, reporting the affine offset rather than absorbing it.
7. Showed the ε -network's error is flat across hierarchy levels while EM-BP's is concentrated by 2–4 orders of magnitude.
8. Reproduced P1's mismatched-oracle result in diffusion: 16 of 16 cells.
9. Corrected two of my own claims — the filtering construction and a false novelty claim — and recorded one clean negative and one confound.
10. Completed the replicated architecture comparison, which qualifies the Layer-5 headline and then partly un-qualifies it.

Test suite: 50 $\rightarrow$ 101, all passing. `main` untouched at cca0018.